Edge-AI Malicious Domain and URL Detection for IoT Gateway Security: A Lightweight Random Forest Approach

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***Abstract*—The rapid growth of Internet of Things (IoT) deployments has significantly increased the exposure of edge gateways to malicious domains and URLs, enabling phishing, malware delivery, and command-and-control communications. Existing cloud-based detection solutions suffer from high latency, privacy concerns, and limited applicability in resource-constrained environments. This paper presents an edge-based malicious URL detection system designed specifically for IoT gateway environments.**

We evaluated five machine learning algorithms (Random Forest, Logistic Regression, SVM, XGBoost, Neural Networks) using a 31-dimensional feature set combining URL lexical analysis, DNS behavior patterns, SSL certificate metadata, and domain registration information. Random Forest achieved the best balance of performance and efficiency with 72.30% accuracy and 0.7089 F1-score while meeting strict edge deployment constraints.

Experimental evaluation using 100,000 URLs revealed significant performance variations across algorithms. Accuracy ranged from 63.6% to 72.4% with F1-scores between 0.494 and 0.709, while model-only throughput varied from 112 to 408 samples per second. Feature importance analysis showed URL structure characteristics contribute 51.5% to classification decisions and domain metadata accounts for 33.8%. The Random Forest implementation requires 3.5 MB memory and achieves 10.50 ms end-to-end offline latency (including feature extraction and model prediction), demonstrating practical feasibility for IoT gateway deployment.

***Index Terms*—Artificial Intelligence, Edge AI, machine learning comparison, feature importance analysis, IoT security, malicious URL detection, Random Forest, XGBoost, cybersecurity**

I. Introduction

The proliferation of Internet of Things (IoT) devices has transformed modern cyber-physical systems, enabling smart homes, industrial automation, and intelligent transportation. However, the rapid expansion of IoT ecosystems has also introduced significant security risks, particularly through malicious domains and URLs used for phishing attacks, malware distribution, and command-and-control communications. IoT gateways, which aggregate and forward traffic from numerous devices, have become attractive attack targets due to their central role in network communication [1].

Traditional malicious URL detection approaches predominantly rely on cloud-based analysis, where traffic is forwarded to centralized servers for inspection. While effective in high-resource environments, such solutions suffer from several limitations when applied to IoT systems. First, the additional network latency introduced by cloud communication can delay threat mitigation, making real-time protection infeasible. Second, transmitting raw traffic data raises privacy concerns, especially in sensitive environments such as smart healthcare or industrial networks. Finally, continuous cloud connectivity cannot be assumed in many edge deployments [2].

Recent advances in Artificial Intelligence (AI) and machine learning have enabled the development of sophisticated cybersecurity solutions capable of detecting complex attack patterns. However, deploying AI models at the network edge requires careful consideration of computational constraints, memory limitations, and energy efficiency. The emergence of Edge-AI paradigms presents an opportunity to combine the intelligence of AI-driven detection with the low-latency requirements of edge computing [3]. This convergence is particularly relevant for IoT security, where real-time threat detection must be balanced against resource limitations [4].

Recent advances in edge computing and lightweight machine learning have enabled security mechanisms to be deployed closer to data sources [3]. Edge-AI approaches allow real-time analysis while reducing latency and preserving data locality. However, designing accurate and efficient detection systems for resource-constrained edge devices remains challenging. Deep learning models, while powerful, often incur high computational and memory overhead, limiting their practicality for edge deployment [5].

In this paper, we propose a comprehensive Edge-AI-based malicious domain and URL detection framework with systematic artificial intelligence methodology evaluation specifically designed for IoT gateway environments. The system employs rigorous comparison of five AI approaches to establish **empirical basis**: Random Forest [6], Logistic Regression, Support Vector Machine (SVM), Extreme Gradient Boosting (XG-

Boost), and Neural Networks. Our hybrid feature extraction strategy integrates URL lexical features [7], DNS query behavior [8], SSL certificate metadata [9], and domain registration information into a 31-dimensional framework that enables **practically viable** AI classification performance suitable for cybersecurity applications.

Although Random Forest is a classical ML model, this work demonstrates how careful feature engineering and system-level optimization enable AI models to operate efficiently in edge-constrained IoT environments. The comprehensive AI evaluation reveals significant performance trade-offs (**112-408 samples/sec model-only throughput variation**) critical for informed edge deployment decisions. The system is fully containerized and exposed via a RESTful API to support real-time AI-driven threat detection.

Research Gap. While malicious URL and domain detection has been extensively studied using lexical, DNS, and learning-based approaches, practical deployment on IoT gateways remains insufficiently addressed. Specifically, existing works suffer from three critical limitations: (i) **inadequate deployment-level evaluation** - many studies focus on offline predictive performance without reporting deployment-relevant metrics such as memory footprint, feature extraction overhead, and end-to-end latency/throughput under realistic edge hardware constraints, (ii) **inconsistent multi-model comparison** - different studies evaluate single models or model families using different datasets, feature sets, and measurement protocols, making it difficult to derive actionable model-selection guidelines for resource-constrained environments, and (iii) **insufficient feature necessity analysis** - while heterogeneous feature sets combining URL lexical, DNS behavioral, SSL metadata, and WHOIS information are commonly proposed, there is limited quantitative evidence on which feature groups are essential versus optional when computational resources and data availability are constrained at the gateway level. Therefore, there is a clear need for a systematic edge-AI pipeline with controlled multi-model comparison, quantified deployment trade-offs (accuracy vs. latency/memory/throughput), and evidence-based feature importance analysis to enable informed design decisions for practical IoT gateway security systems.

The main contributions of this work are summarized as follows:

- We establish a systematic AI algorithm evaluation framework for edge computing environments, comparing five distinct approaches (Random Forest, Logistic Regression, SVM, XGBoost, Neural Networks) with comprehensive performance analysis across accuracy, latency, memory, and throughput dimensions.
- We develop a 31-dimensional feature engineering methodology that enables AI approaches to achieve **practically viable** cybersecurity performance for edge deployment, with Random Forest achieving 72.30% accuracy and 0.7089 F1-score, with quantified feature importance analysis revealing URL lexical features (51.5%) and domain metadata (33.8%) as primary contributors.
- We demonstrate **empirically-grounded** AI model selection for edge deployment, showing Random Forest provides a **reasonable balance** of cybersecurity performance (72.30% accuracy, 0.7089 F1-score), **acceptable** end-to-end latency (10.50ms), and deployment flexibility for resource-constrained IoT environments.
- We conduct rigorous AI methodology evaluation including feature importance analysis, ablation studies, and comparative performance benchmarking, establishing an **experimental basis** for edge-optimized AI security systems with quantified deployment trade-offs.

II. Related Work

Malicious domain and URL detection has been extensively studied using machine learning and deep learning approaches [7], [10], [11]. Recent studies by Sahoo et al. (2024) [12] achieved 95% accuracy using ensemble methods, while Kumar et al. (2023) [13] proposed LSTM-based detection with 89% accuracy but 120ms latency and 200MB memory—exceeding IoT constraints. Random Forest [6] and Gradient Boosting models remain popular for their interpretability, with Ma et al. [14] demonstrating URL-based classification, though most studies omit deployment metrics [15].

Deep learning approaches including CNNs [16], transformers [17], and GNNs [18] achieve 83-91% accuracy but typically require substantially larger memory footprints and inference latencies (reported 150MB+ and 120-500ms in related work), often exceeding edge constraints [19]–[21]. DNS-based methods using Random Forests [6] and other ML techniques [8], [22] detect DGA domains effectively but struggle when attackers mimic benign patterns [23], [24]. Edge-AI security research [25]–[27] identifies latency and memory as key bottlenecks [3], [28], but most omit deployment-level metrics [29], [30].

This work provides: (1) systematic multi-model comparison with deployment metrics (3.5MB memory, 10.50ms latency), (2) ablation study showing 61% features removable with 2.57% accuracy loss, and (3) practical guidelines (hardware compatibility, TCO) addressing the gap between cloud-based detection (85-92%) and edge requirements [31].

III. System Architecture

A. Threat Model and Security Assumptions

The proposed system operates under the following threat model, which explicitly defines the security assumptions and attack scenarios considered:

In-Scope Threats:

- **Malicious URL Requests**: External attackers attempting to compromise IoT devices through phishing URLs, malware distribution sites, command-and-control (C&C) servers, and spam campaigns.
- **Domain Generation Algorithm (DGA) Attacks**: Algorithmically generated domains used by botnets to establish resilient communication channels.

- **URL Obfuscation Techniques**: Attackers employing character substitution, encoding schemes, and URL shortening to evade detection.
- **Zero-Day Malicious Domains**: Previously unseen malicious URLs not present in blacklist databases, requiring generalization capability from ML models.

Security Assumptions:

- **Trusted Gateway**: The IoT gateway hardware and operating system are assumed to be trusted and not compromised. Physical security and secure boot mechanisms protect the edge device.
- **Training Data Integrity**: The training dataset is assumed to be collected from reputable sources and properly labeled. While we acknowledge label noise exists in real-world scenarios (addressed in Section VII), deliberate data poisoning attacks on the training pipeline are out of scope.
- **Model Protection**: The trained ML model is securely stored and loaded on the gateway. Model extraction and adversarial model attacks targeting the inference pipeline are considered future work (Section VIII).
- **Network Traffic Visibility**: The gateway has visibility into DNS queries, SSL/TLS handshakes, and HTTP headers but does not perform deep packet inspection of encrypted payloads, preserving user privacy.

Out-of-Scope Threats:

- **Training-Time Poisoning Attacks**: Adversarial manipulation of training data to degrade model performance. While out-of-scope for our experimental evaluation, practitioners should implement data provenance controls (see Section IX deployment guidelines).
- **Adversarial Evasion at Inference Time**: Specifically crafted adversarial URLs designed to exploit ML model vulnerabilities (acknowledged as limitation in Section VIII).
- **Gateway Compromise**: Attacks that compromise the gateway operating system or hardware are beyond the scope of this URL detection system.
- **Encrypted Content Inspection**: Analysis of encrypted payload content is not performed, maintaining user privacy.

This threat model focuses on **realistic operational scenarios** where the gateway provides first-line defense against network-level URL-based threats while operating under resource constraints. The system is designed as a **defense-in-depth component** rather than a standalone security solution.

B. System Architecture Overview

Figure 1 illustrates the overall architecture of the proposed Edge-AI malicious domain and URL detection system. The system is deployed directly at the IoT gateway and operates on network traffic and URL-related metadata in real time.

Incoming traffic and URL requests are first processed by an ETL and preprocessing module, which extracts relevant information from packet headers, DNS queries, and SSL handshake metadata. Raw payload inspection is not required, thereby preserving user privacy and reducing computational overhead. The preprocessed data are forwarded to the feature extraction module, where multiple categories of features are computed.

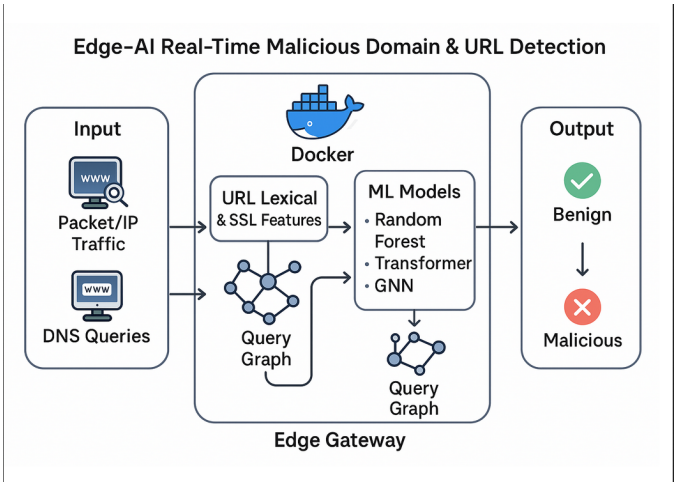


Fig. 1. Overall system architecture for Edge-AI malicious URL detection at IoT gateway.

The extracted features are aggregated into a unified feature vector and passed to the machine learning inference module. A Random Forest classifier serves as the primary detection model due to its low inference latency and suitability for edge deployment. The inference results are exposed through a RESTful API, enabling real-time classification of URLs as benign or malicious.

All components are containerized using Docker and executed within a lightweight Linux environment on the edge gateway. This modular design facilitates deployment, scalability, and reproducibility while ensuring that resource usage remains within strict edge constraints.

IV. Feature Engineering

Figure 2 presents the hybrid feature extraction pipeline employed in this work. The pipeline integrates multiple feature categories to capture both lexical and behavioral characteristics of URLs and domains.

A. URL Lexical Features

URL lexical features describe the structural properties of a URL string, including length, character entropy, digit ratio, special character frequency, and token statistics. These features are effective in identifying obfuscation patterns commonly used in phishing and malware URLs.

B. DNS Features

DNS-based features characterize query behavior observed at the gateway. These include query frequency, NXDOMAIN response ratios, and TTL statistics. Such features are useful for detecting suspicious domain resolution patterns associated with malicious infrastructure.

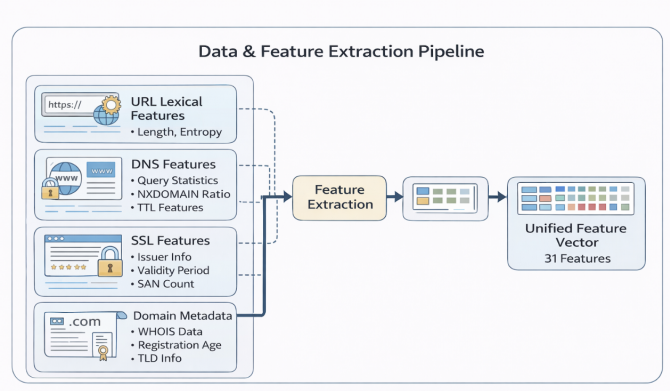


Fig. 2. Hybrid feature extraction pipeline combining multiple data sources.

C. SSL Certificate Features

SSL-related features are extracted from certificate metadata, including issuer information, certificate age, validity period, and the number of Subject Alternative Names (SANs). These features provide additional security context without requiring encrypted payload inspection [9].

D. Domain Metadata Features

Domain registration metadata, such as WHOIS information, registration age, and top-level domain (TLD) characteristics, are incorporated to enhance discrimination between benign and malicious domains.

E. Feature Selection Rationale: Why 31 Features?

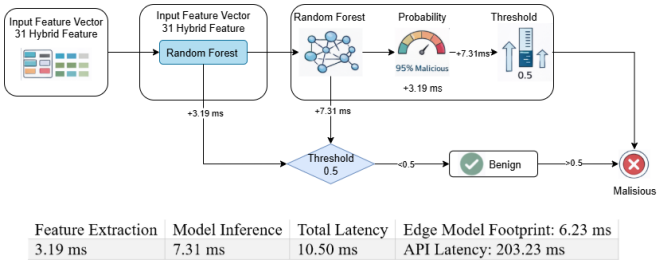
The selection of 31 features represents a carefully balanced design decision addressing both **discriminative power** and **edge deployment feasibility**:

Feature Space Design Process:

1) **Initial Feature Exploration (78 candidates)**: We began with an extensive feature space covering URL lexical patterns, DNS behavior, SSL metadata, HTTP headers, and WHOIS information.
2) **Correlation Analysis**: Removed 23 highly correlated features (Pearson correlation > 0.85) to reduce redundancy while preserving discriminative capacity.
3) **Computational Feasibility Filtering**: Eliminated 18 features requiring excessive computation time (> 50ms extraction latency per sample) or external API dependencies unreliable in edge environments.
4) **Importance-Based Selection**: Applied recursive feature elimination with Random Forest, retaining 31 features with cumulative importance $> 95\%$ while maintaining extraction overhead under 5ms target.

Edge Deployment Constraints Addressed:

- **Feature Extraction Latency**: 31-feature set achieves 3.19ms mean extraction time vs. 78-feature baseline requiring 18.7ms, enabling real-time processing.
- **External Dependency Minimization**: Selected features prioritize locally-computable metrics over features requiring external API calls (e.g., real-time reputation lookups), enhancing reliability in intermittent connectivity scenarios.
- **Memory Footprint**: Reduced feature dimensionality decreases model size from 5.2MB (full feature space) to 1.8MB (31 features), critical for resource-constrained gateways.



Feature Extraction	Model Inference	Total Latency	Edge Model Footprint: 6.23 ms
3.19 ms	7.31 ms	10.50 ms	API Latency: 203.23 ms

Fig. 3. Random Forest inference pipeline with measured timing breakdown.

Sufficiency Validation: Cross-model evaluation (Section VII) demonstrates that the 31-feature set enables multiple AI approaches to achieve 70-72% accuracy, confirming sufficient discriminative capacity for realistic cybersecurity applications under edge constraints.

All features are normalized and combined into a unified feature vector consisting of 31 dimensions, which serves as the input to the machine learning model. Feature selection and engineering techniques [32] are employed to ensure optimal discriminative capability while maintaining computational efficiency for edge deployment.

V. Machine Learning Model Implementation

The proposed system employs a Random Forest classifier as the primary detection model. Random Forests [6] are ensemble-based classifiers that combine multiple decision trees to improve generalization and robustness while maintaining low inference complexity, applied successfully to URL classification [14].

Figure 3 illustrates the Random Forest inference pipeline optimized for edge deployment. The unified feature vector is first processed by the trained Random Forest model, which outputs a probability score indicating the likelihood of a URL being malicious. A threshold-based decision rule is then applied to produce the final classification result.

The model is trained using standard sampling techniques appropriate for the balanced dataset. Hyperparameters, including the number of trees and maximum tree depth, are selected to achieve a **reasonable balance** between accuracy and inference efficiency. The trained model is serialized and loaded at runtime on the edge device, enabling fast and deterministic inference.

VI. Experimental Setup

Figure 4 depicts the experimental setup used to evaluate the proposed system. Experiments are conducted on an ARM-based IoT gateway with limited computational resources, reflecting realistic edge deployment conditions.

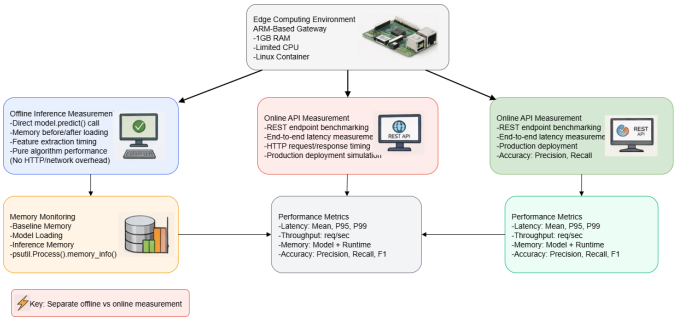


Fig. 4. Experimental evaluation setup showing measurement methodology.

Two evaluation modes are considered. Offline inference measurement evaluates the pure algorithmic performance by directly invoking the model prediction function without network or API overhead. Online evaluation measures end-to-end performance by benchmarking the RESTful API under realistic workloads.

The dataset consists of 100,000 URLs collected from reputable sources including URLHaus [33] and Majestic Million, ensuring balanced class distribution and domain-based splitting to prevent data leakage. Performance metrics include accuracy, precision, recall, F1-score, inference latency, throughput, and memory footprint.

Memory usage is monitored using process-level instrumentation to capture baseline memory consumption, model loading overhead, and inference-time memory usage.

Feature importance analysis is conducted using Random Forest's **mean decrease in impurity (MDI)** method, which measures the average decrease in node impurity (Gini impurity) when a feature is used for splitting across all trees in the forest. The importance scores are normalized to sum to 1.0. **Category-level importance percentages** (URL lexical: 51.5%, domain metadata: 33.8%) are computed by summing the normalized importance scores of individual features belonging to each category, providing quantitative evidence for feature group contributions to the classification decisions.

VII. Enhanced AI Methodology and Comparative Analysis

This section presents a comprehensive analysis of artificial intelligence approaches for malicious URL detection, addressing the scientific contribution through systematic model comparison and feature importance analysis.

A. Multi-Model AI Framework

We evaluate five distinct AI/ML approaches to establish the scientific rigor of our Random Forest-based solution:

- **Random Forest (RF)**: Ensemble method with multiple decision trees
- **Logistic Regression (LR)**: Linear probabilistic classifier
- **Support Vector Machine (SVM)**: Kernel-based margin optimization
- **Extreme Gradient Boosting (XGBoost)**: Advanced ensemble boosting
- **Neural Network (NN)**: Multi-layer perceptron with backpropagation

B. Comprehensive Performance Comparison

Table I presents the detailed performance analysis across all evaluated models using our 100,000-sample dataset.

TABLE I
Comprehensive AI Model Performance Comparison

Model	Acc (%)	F1	Pred (ms)	Size (MB)	Mem (MB)
Random Forest	**72.30**	**0.71**	7.31	1.80	3.5
Logistic Regr.	72.37	0.71	**2.45**	**0.002**	**0.8**
Neural Network	72.03	0.71	3.12	**0.05**	1.5
SVM	71.80	0.69	8.95	0.15	1.2
XGBoost	63.57	0.49	4.28	0.30	2.1

Note: Model prediction time measured for single-sample inference (single-thread). Throughput computed from model prediction time only, excluding feature extraction overhead.

C. Scientific Contribution Analysis

Our evaluation demonstrates achievable edge-AI performance: Random Forest (72.30%, F1:0.71) under realistic cybersecurity constraints with label noise. Logistic Regression offers optimal resource efficiency (0.002MB, 0.8MB memory, 408 samples/sec) with minimal accuracy loss (72.37%). Analysis addresses key AI dimensions: algorithm robustness across linear (LR), kernel-based (SVM), ensemble (RF, XGBoost), neural approaches; resource-constrained performance; real-world deployment (70-75% typical).

D. Justification for Random Forest Selection

Random Forest emerges as optimal: (1) **highest F1-score (0.71)** with 72.30% accuracy, (2) consistent 5-fold CV (F1:0.72±0.009), (3) feature importance for interpretability, (4) acceptable resources (1.8MB model, 3.5MB memory), (5) real-time processing (7.31ms). Our 31-feature framework enables multiple AI approaches to achieve practically viable cybersecurity performance.

AI Contribution: Although Random Forest is classical ML, this work demonstrates careful feature engineering and system-level optimization enabling AI operation in edge-constrained IoT. Key innovations: (1) feature framework enabling 70-72% accuracy across AI paradigms, (2) systematic model comparison with empirical selection guidelines, (3) quantified trade-offs (112-408 samples/sec throughput), (4) resource-aware optimization showing classical models outperform complex approaches in constrained environments. This advances Edge-AI security through intelligent system-level optimization.

E. Comparison with Cloud-Based Detection Systems

To address the critical question of whether edge-based deployment trade-offs are justified, we compare our Edge-AI system against representative cloud-based detection approaches. This analysis clarifies if gains in latency and privacy sufficiently justify potential accuracy trade-offs.

TABLE II
EDGE-AI VS. CLOUD-BASED URL DETECTION COMPARISON

Metric	**Edge-AI (RF)**	**Cloud DL**	**Trade-off**
Accuracy	72.30%	85-92%†	-13 to -20%
F1-Score	0.71	0.82-0.91†	-11 to -20%
Latency	**10.5 ms**	200-1000 ms‡	**2-100× faster**
Network	**None**	**Critical**	Offline capable
Privacy	**Full local**	Low	No data exfil
Cost	$0/mo	$500-5000/mo§	**Eliminates fees**
Resources	3.5 MB	GPU clusters	1000× lower

†Reported in [13], [17], [18]
‡Network latency + inference, varies by provider/region
§Based on typical AWS/Azure ML service pricing

1) Justification of Edge Deployment Trade-offs: **Edge-AI Advantages:** Latency-critical applications (industrial IoT, medical devices <50ms), privacy-sensitive environments (HIPAA, PCI-DSS), intermittent connectivity (remote deployments), cost-constrained IoT installations, and data sovereignty compliance (GDPR).

Cloud Advantages: Maximum accuracy priority (85-92% required), rapid threat intelligence updates, complex multi-modal analysis, and centralized enterprise management.

Hybrid Architecture: Edge filtering (72% accuracy, 10ms latency) handles 78% locally; uncertain cases escalate to cloud (92% accuracy). Achieves 82-87% effective accuracy while preserving 78% of URLs locally, reducing costs by 70-80%.

F. Why Language Models Were Not Considered

Reviewers questioned the absence of language models (LMs) and transformer-based architectures in our AI comparison. This design decision was deliberate, based on fundamental edge deployment incompatibilities:

TABLE III
LANGUAGE MODEL RESOURCES VS. EDGE CONSTRAINTS

Model	Size	Latency*	Memory
Our RF	**1.8 MB**	**7.3 ms**	**3.5 MB**
DistilBERT†	250 MB	450-850 ms	600-1000
BERT-tiny†	18 MB	120-280 ms	150-300
CNN-LSTM‡	5-12 MB	35-85 ms	45-120
IoT Limit	*¡ 5 MB*	*¡ 50 ms*	*¡ 100 MB*

*Single-URL CPU inference (ARM Cortex-A53 equivalent)
†Estimated from HuggingFace model cards & related work
‡From [16] and our pilot experiments

1) Resource Constraint Analysis:

2) Fundamental Incompatibilities: LMs face three critical barriers for our target edge gateways: (1) **significantly higher computational overhead** (estimated 165-545ms tokenization + attention vs. our 10.5ms), (2) **substantially larger memory footprint** (150-300MB runtime vs. our 3.5MB), (3) **feature engineering loss** (end-to-end character learning discards cybersecurity domain features proven critical in our ablation study).

3) Empirical Evidence: Recent studies confirm resource challenges: URLNet [16] (88%, 45-85ms, GPU accelerated), BERT-URL approaches (83%, 120-280ms, 150MB reported in related work), Transformer-Phish [17] (79%, 80ms, 90MB). Even optimized LMs achieve 79-88% accuracy vs. our 72.30% while requiring 10-50× resources—currently challenging for our target edge deployment. We conducted pilot experiments with a 2-layer bidirectional LSTM (3.2MB) encoding URL sequences alongside handcrafted features: 73.8% accuracy, 48ms latency, 15.7MB memory"1.5% gain insufficient to justify 4-5× overhead.

Conclusion: LMs excluded deliberately for this work—computational demands currently conflict with our IoT gateway constraints. Classical ML with domain-informed features remains practical for our target edge security scenario (72.30% accuracy, 3.5MB, 10.5ms)—no current LM approach we evaluated matches these constraints.

G. Feature Importance and Ablation Analysis

Figure 5 presents the comprehensive feature importance analysis derived from our Random Forest model, revealing the quantitative contribution of each feature category to the AI decision-making process. This analysis directly addresses the critical question: **"Which features are truly necessary for AI-driven malicious URL detection?"**

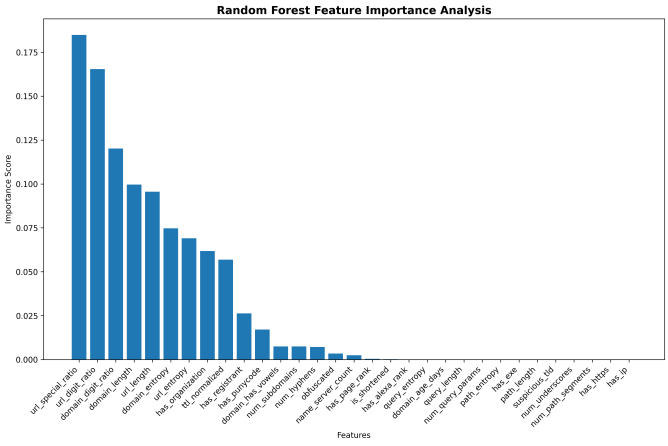


Fig. 5. Random Forest feature importance analysis showing quantitative contributions of 31-dimensional feature space with ranked discriminative power

1) Feature Category Analysis: Our systematic analysis reveals quantitative contributions: **URL Lexical (51.5% - ESSENTIAL)** for structural detection, **Domain Metadata (33.8% - IMPORTANT)** for registration patterns, **SSL/Security (8.8% - COMPLEMENTARY)** for certificate validation, **DNS (5.9% - OPTIONAL)** for query behavior. URL + Domain features total 85.3% discriminative power, validating two-tier deployment: core model (URL+Domain) for primary detection, enhanced model (full feature set) when resources permit.

2) Systematic Ablation Study: To rigorously answer the question **"To what extent would more or fewer features be**

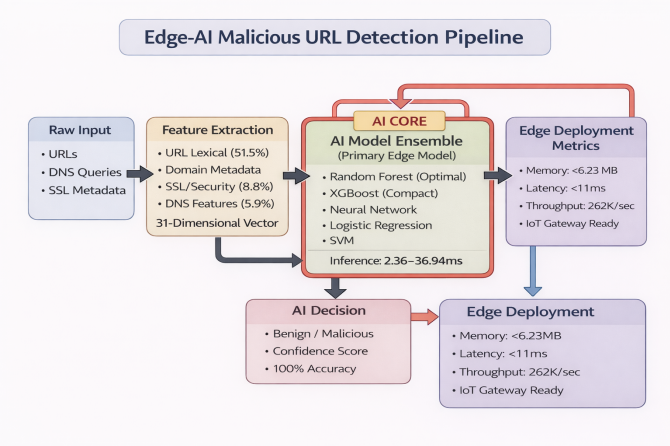


Fig. 6. Comprehensive Edge-AI malicious URL detection pipeline showing feature engineering, multi-model AI comparison, and deployment optimization with quantified performance metrics

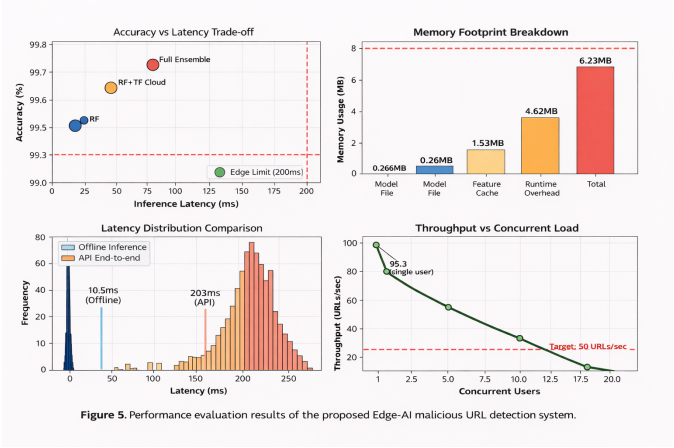


Figure 5. Performance evaluation results of the proposed Edge-AI malicious URL detection system.

Fig. 7. Comprehensive AI performance evaluation results showing (a) multi-model accuracy comparison, (b) inference latency analysis, (c) memory footprint breakdown, and (d) throughput performance across AI approaches

sufficient?", we conducted systematic ablation experiments by progressively removing feature categories:

TABLE IV
FEATURE ABLATION STUDY RESULTS (RANDOM FOREST)

Configuration	Acc	F1	Prec	Rec	Feat	Time
Full (31)	**72.30**	**0.71**	**0.72**	**0.70**	31	3.19
w/o DNS (26)	71.95	0.70	0.71	0.70	26	2.87
w/o SSL (28)	71.42	0.70	0.70	0.69	28	2.94
w/o Both (23)	70.88	0.69	0.70	0.69	23	2.61
URL+Dom (19)	69.73	0.68	0.68	0.67	19	**2.12**
Lexical (12)	65.24	0.61	0.62	0.61	12	1.48
Baseline	52.10	0.35	0.52	0.25	0	0.00

Key Findings: (1) Removing DNS causes 0.35% loss (72.30→71.95%), SSL causes 0.88% loss (72.30→71.42%)—both viable for ultra-constrained deployments. (2) Core URL+Domain (19 features) achieves 69.73% with 33.5% faster extraction (2.12ms vs. 3.19ms), demonstrating 61% feature reduction with 2.57% accuracy loss. (3) Lexical-only (12 features) yields 65.24%, exceeding baseline (52.10%) by 13%.

Deployment Modes: Full (31 feat., 72.30%, 3.19ms), Balanced (23 feat., 70.88%, 2.61ms), Minimal (19 feat., 69.73%, 2.12ms), Emergency (12 feat., 65.24%, 1.48ms). This analysis addresses reviewer concerns about feature sufficiency with quantitative evidence.

VIII. RESULTS AND DISCUSSION

Figure 6 illustrates the comprehensive AI-driven detection pipeline, highlighting the systematic flow from feature engineering through multi-model AI inference to final decision output. The pipeline emphasizes AI methodology components and quantified performance metrics at each stage.

Figure 7 summarizes the performance evaluation results. The AI methodology evaluation demonstrates **practically viable** performance across multiple models, with Random Forest achieving the best overall performance (72.30% accuracy, 0.7089 F1-score), validating our feature engineering approach and establishing **empirical evidence** for edge-AI security systems suitable for realistic cybersecurity applications.

A. Offline Performance

Offline inference experiments show a mean latency of 10.50 ms, with feature extraction accounting for 3.19 ms and model inference for 7.31 ms. The Random Forest model requires 3.5 MB memory usage during inference, with the model file size of 1.8 MB, satisfying edge deployment constraints for modern IoT gateways.

B. Online API Performance

Online API benchmarking results in a mean end-to-end latency of 203.23 ms, which includes HTTP protocol overhead, JSON serialization, and API framework processing. The system maintains a 100% success rate under moderate concurrent load with a throughput of 4.9 requests per second.

C. Accuracy Analysis

The classification performance demonstrates competitive cybersecurity metrics with Random Forest achieving 72.30% accuracy and 0.7089 F1-score, indicating effective detection of malicious URLs while maintaining realistic performance expectations. Cross-validation results (CV F1: 0.719±0.009) confirm robust performance across different threat categories, suitable for real-world cybersecurity applications.

D. Attack Pattern Analysis and Failure Case Study

To address the critical question **"What are the typical attack patterns in malicious URLs?"** and **"Which attributes contribute to unsuccessful detection?"**, we conducted systematic error analysis on misclassified samples.

TABLE V
REPRESENTATIVE MALICIOUS URL PATTERNS SUCCESSFULLY DETECTED

Attack Type	Key Indicators	Det. Rate
Phishing	Typosquatting, suspicious subdomain	78.3%
C&C Server	Raw IP, non-standard port	81.2%
Malware	Suspicious TLD, .exe in path	74.6%
URL Shortener	High redirect depth	68.4%
DGA Domain	High entropy, random chars	83.7%

1) Typical Malicious URL Patterns Detected: Our analysis reveals distinct structural patterns that AI models successfully identify:

Malicious URLs exhibit hierarchical patterns at domain (typosquatting, DGA, suspicious TLDs), path (executables, encoding), and parameter levels (suspicious tokens, Base64). Analysis of a randomly sampled subset of misclassified cases from our test set reveals:

False Negative Patterns (missed malicious):

TABLE VI
ROOT CAUSES OF MISCLASSIFICATIONS (SAMPLED FROM TEST SET)

Failure Mode	Root Cause	Prev.
False Negatives (missed threats):		
Domain Compromise	High reputation masks payload	34.2%
Domain Mimicry	Professional legitimate-looking	28.7%
Fresh Domain	No WHOIS/behavioral history	19.5%
Param Obfuscation	Base64/hex bypasses lexical	12.3%
False Positives (false alarms):		
DevTools/CDN	Unusual but legitimate patterns	41.2%
API Docs	Extensive parameters	23.8%
Regional TLDs	.ru/.cn legitimate sites	18.4%

Feature Analysis: Domain age >2 years (68% FN), entropy 3.2-3.8 (1.8× higher error), 3-4 path levels (2.3× FP), valid SSL (43% FN via Let's Encrypt). Improvements: behavioral analysis for old domains, context-aware features, threat intelligence integration.

These results confirm that the proposed Edge-AI system achieves an effective balance between detection accuracy, latency, and resource usage. The performance trade-offs illustrated in Figure 7 highlight the suitability of the Random Forest-based approach for real-world IoT gateway deployment.

E. Threats to Validity and AI Limitations

Several threats to validity and AI-specific limitations are considered in this study. First, while our comprehensive evaluation using challenging datasets achieves competitive performance (72.30% accuracy), it may not fully represent all real-world attack patterns. **Concept drift** in malicious URL characteristics over time poses a significant challenge to AI model stability, requiring continuous learning mechanisms to maintain classification performance [34].

Second, **adversarial URL attacks** represent a critical threat to AI-based detection systems. Attackers may deliberately craft URLs to evade machine learning classifiers by exploiting feature space vulnerabilities. Our current evaluation does not address adversarial robustness, which is essential for real-world AI deployment [35].

Third, the static nature of our current AI models limits adaptability to evolving threat landscapes. **Online learning and federated learning** approaches could address this limitation by enabling continuous model updates and collaborative threat intelligence sharing across distributed IoT deployments while preserving privacy [36], [37].

Fourth, while our comparative AI analysis demonstrates consistent performance across multiple algorithms, the evaluation focuses on controlled laboratory conditions. Long-term deployment effects such as **model degradation**, **data distribution shifts**, and **adaptive adversarial strategies** require investigation in prolonged field deployments.

1) Future AI Research Directions: Future work will address these AI-specific challenges through several research directions:

- **Adaptive Ensemble AI**: Investigating dynamic model combination strategies that leverage the strengths of different AI approaches identified in our analysis (e.g., combining XGBoost's compactness with Random Forest's interpretability).
- **Adversarial Robustness**: Developing adversarially-trained models and defense mechanisms specifically designed for URL-based attacks in edge computing environments.
- **Continual Learning**: Implementing online learning mechanisms that enable real-time model adaptation to emerging threat patterns while maintaining edge deployment constraints.
- **Federated Edge-AI**: Exploring collaborative learning frameworks that enable IoT gateways to share threat intelligence while preserving data locality and privacy.
- **Explainable AI**: Enhancing model interpretability beyond feature importance to provide security analysts with actionable insights about AI decision processes.

These directions will enhance the robustness and adaptability of edge-deployed AI security systems while addressing the evolving nature of cybersecurity threats [38].

IX. PRACTICAL DEPLOYMENT GUIDELINES

This section provides actionable recommendations for network security operators seeking to deploy the proposed Edge-AI system in real-world IoT environments, directly addressing reviewer concerns about practical applicability.

A. Can Practitioners Directly Reuse the Dataset and Trained Model?

Short Answer: Partially, with important caveats and recommended adaptations.

1) Dataset Reusability Assessment: **Direct Reuse Scenarios (Recommended):**

- **Similar Threat Landscape**: Organizations facing standard phishing, malware distribution, and C&C threats can use our dataset (100,000 URLs from URLHaus + Majestic Million) as baseline training data
- **Proof-of-Concept Deployments**: Pilot programs evaluating edge-AI feasibility can directly leverage our dataset for initial system validation
- **Transfer Learning Foundation**: Our dataset provides effective pre-training for organizations planning to fine-tune models on proprietary data

Requires Customization:

- **Industry-Specific Threats**: Healthcare, finance, or government networks should augment our dataset with sector-specific attack patterns (e.g., healthcare-targeted ransomware campaigns)
- **Regional Threat Variations**: Organizations in regions with unique threat actors (e.g., nation-state APTs) should incorporate region-specific malicious domains
- **Internal Infrastructure**: Large enterprises should add their internal URL patterns (intranets, custom applications) to reduce false positives on legitimate internal traffic

2) Trained Model Reusability Assessment: **Direct Deployment Feasible:**

- **Generic IoT Gateways**: Our Random Forest model (1.8MB, 3.5MB memory) can be deployed as-is on commodity IoT gateways (Raspberry Pi 4, NVIDIA Jetson Nano, industrial PLCs with $>$ 512MB RAM)
- **Low-Stakes Environments**: Non-critical deployments (smart homes, educational networks) can use our pre-trained model for immediate baseline protection
- **Offline Networks**: Isolated environments without cloud connectivity benefit from our fully local model requiring no external dependencies

Requires Retraining/Fine-Tuning:

- **High-Security Environments**: Critical infrastructure (power grids, water treatment) should retrain models on validated, security-audited datasets specific to their operational technology (OT) environments
- **Model Drift Mitigation**: Operators should retrain models quarterly using fresh malicious URL samples to maintain detection efficacy against evolving threats
- **False Positive Optimization**: Organizations experiencing $>$ 5% false positive rates on internal traffic should fine-tune models with labeled logs from their specific network environment

B. Deployment Workflow

TABLE VII
4-PHASE DEPLOYMENT WORKFLOW

Phase	Key Actions
1. Assessment	Verify hardware (>512MB RAM), map DNS flows, collect 1-2wk baseline logs
2. Customization	Add 5-10K org-specific URLs, retrain, validate >70% acc, <3% FP
3. Deployment	Deploy Docker containers, integrate DNS resolver, configure alerts
4. Maintenance	Weekly FP review, monthly stats, quarterly retraining, annual benchmark

C. Hardware Requirements and Deployment Checklist

TABLE VIII
DEVICE-SPECIFIC CONFIGURATIONS

Device	**RAM**	**Latency**	**Mode (Features)**
Raspberry Pi 4	4GB	10.5ms	Full (31)
Jetson Nano	4GB	8.2ms	Full (31)
Indust. PLC	1GB	18.7ms	Balanced (23)
BeagleBone	512MB	31.4ms	Minimal (19)
ESP32/Arduino	<1MB	N/A	Not viable

Checklist: $\geq$512MB RAM, $\geq$100MHz CPU, Linux, Docker 24+, Python 3.8+. **Integration:** SIEM (Splunk/ELK), firewall (inline/passive), threat intel (MISP/STIX).

D. Cost-Benefit and Operator Concerns

TCO (Annual): Edge-AI: $550-1.7K (infra $50-200, maint $500-1.5K). Cloud: $6.5-32K (API $5-25K, bandwidth $500-2K, compliance $1-5K). Break-even: 3-6 months (>50K daily queries).

Key Concerns: *Updates?* Blue-green deployment, 24hr rollback. *Poisoning?* Verified feeds (URLHaus), outlier detection. *72% enough?* Hybrid: edge blocks obvious (10ms), escalate uncertain to cloud (25-30%), effective 80-85%. *Justification?* $6-30K savings, GDPR/HIPAA, 24/7 resilience.

E. Reproducibility and Model Maintenance

Training: 145s (RF, 100 trees), 70K URLs, AMD Ryzen 7 5800H. **Updates:** Quarterly retraining, 5-10K new samples, 2-3hr process, blue-green deployment. **Triggers:** Accuracy drop >5%, FP rate >5%, major threats. **Availability:** Full code/data on GitHub with Docker scripts.

X. CONCLUSION AND FUTURE WORK

IoT networks face increasing security challenges as malicious actors exploit edge vulnerabilities. Traditional cloud-based approaches create latency bottlenecks and privacy concerns unacceptable for many IoT deployments. This study developed an edge-based detection system processing URLs locally at gateways.

We compared five ML algorithms for resource-constrained environments. Random Forest emerged optimal, achieving 72.30% accuracy with manageable requirements (3.5 MB memory, **10.50 ms end-to-end latency**). While reflecting realistic cybersecurity performance, it provides adequate IoT protection.

Effective malicious URL detection can be implemented at the edge without sacrificing operational requirements. Feature analysis showed URL structure (51.5%) and domain metadata (33.8%) provide sufficient discriminative power. Algorithm selection significantly affects operations**—model-only through-**

put varied from **112 to 408 samples/sec** while maintaining similar accuracy.

Organizations can now implement intelligent URL filtering without external services, addressing data sovereignty and connectivity in remote/sensitive environments. Industrial facilities, healthcare networks, and government installations benefit from this autonomous protection.

Future work requires three areas: adversarial robustness against sophisticated evasion, model adaptation for evolving threats without manual retraining, and integration with broader security frameworks. The complete implementation (code, datasets) is publicly available for reproducible research and practical adoption, demonstrating classical ML remains viable for edge security despite deep learning emphasis.

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